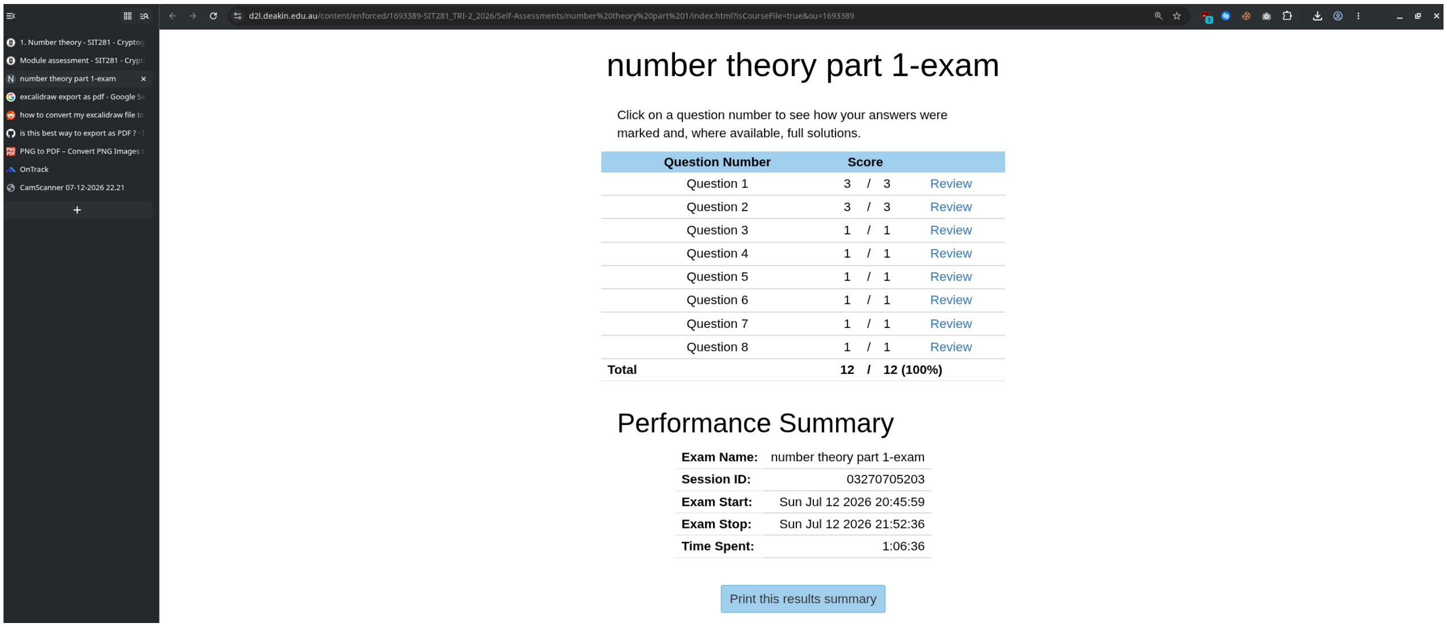
Self-Assessment Evidence

Results Screen



number theory part 1-exam

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	
Question 1	3 / 3	Review
Question 2	3 / 3	Review
Question 3	1 / 1	Review
Question 4	1 / 1	Review
Question 5	1 / 1	Review
Question 6	1 / 1	Review
Question 7	1 / 1	Review
Question 8	1 / 1	Review
Total	**12 / 12 (100%)**	

Performance Summary

Exam Name:	number theory part 1-exam
Session ID:	03270705203
Exam Start:	Sun Jul 12 2026 20:45:59
Exam Stop:	Sun Jul 12 2026 21:52:36
Time Spent:	1:06:36

Print this results summary

Working

① $\gcd(88, 67) \Rightarrow$

	88	67
$88 - 67 \times 1 = 21$	$1 - 0 = 1$	$0 - 1 \times 1 = -1$
$67 - 21 \times 3 = 4$	$0 - 1 \times 3 = -3$	$1 - (-1) \cdot 3 = 4$
$21 - 4 \times 5 = 1$ //	$1 - -3 \times 5 = 16$	$-1 - (4 \times 5) = -21$

$$1 = 88 \times \underbrace{16} + 67 \times \underbrace{-21}$$

$$1 = 88 \times 16 - 67 \times 21$$

② gcd (47, 87) $\Rightarrow$

	47	87
$87 - 47\times 1 = 40$	$0 - 1 = -1$	$1 - 0 = 1$
$47 - 40\times 1 = 7$	$1 - -1 = 2$	$0 - 1\times 1 = -1$
$40 - 7\times 5 = 5$	$-1 - 2\times 5 = -11$	$1 - (-1)\cdot 5 = 6$
$7 - 5\times 1 = 2$	$2 - -11\cdot 1 = 13$	$-1 - 6\times 1 = -7$
$5 - 2\times 2 = 1$ //	$-11 - 13\times 2 = -37$	$6 - -7\times 2 = 20$

$$1 \equiv 47 \times \underbrace{-37} + 87 \times \underbrace{20}$$

③ modular inverse of 89 mod 28

$89 - 28\times 3 = 5 \Rightarrow 89 = 28\times 3 + 5$

gcd (89, 28) $\Rightarrow$

	89
$89 - 28\times 3 = 5$	$1 - 0 = 1$
$28 - 5\times 5 = 3$	$0 - 1\times 5 = -5$
$5 - 3\times 1 = 2$	$1 - -5\times 1 = 6$
$3 - 2\times 1 = 1$	$-5 - 6\times 1 = \underline{\underline{-11}}$ inverse

$-11 + 28 = +17$

$\therefore \Rightarrow$ 17 mod 28 //

(4) inverse of 79 mod 91

$\gcd(79, 91) \Rightarrow$	$\underline{\underline{79}}$
$91 - 79 \times 1 = 12$	$0 - 1 \times 1 = -1$
$79 - 12 \times 6 = 7$	$1 - -1 \times 6 = 7$
$12 - 7 \times 1 = 5$	$-1 - 7 = -8$
$7 - 5 \times 1 = 2$	$7 - -8 = 15$
$5 - 2 \times 2 = 1$	$-8 - 15 \times 2 = -\underline{\underline{38}}$ ← inverse

$-38 + 91 = 53$

$\therefore \Rightarrow 53 \bmod 91$

(5) $x = \overset{a}{\underline{0}} \bmod \overset{m}{\underline{5}}$

$x = \underset{b}{\underline{10}} \bmod \underline{19}\ n$

$\gcd(5, 19) \Rightarrow$	$\underline{\underline{5}}$	$\underline{\underline{19}}$
$19 - 5 \times 3 = 4$	-3	1
$5 - 4 \times 1 = 1$	$1 - -3 = \underset{s}{\underline{4}}$	$\underset{t}{\underline{-1}}$

$x \equiv bms + ant \pmod{m \cdot n}$

$x \equiv 10 \times 5 \times 4 + 0 \pmod{5 \times 19}$

$x \equiv 200 \pmod{95}$

$x \equiv 10 \bmod 95$

$200 = 95 \times 2 + 10$

⑥ $x = \overset{a}{0} \mod \overset{m}{2}$

$x = \underset{b}{26} \mod 29_n$

$\gcd(2, 29) \Rightarrow$

	$\underline{\underline{2}}$	$\underline{\underline{29}}$
$29 - 2\times 14 = 1$	-14_s	1_t

$x \equiv bms + ant \pmod{m\times n}$

$\equiv 26\times 2\times -14 + 0 \pmod{2\times 29}$

$\equiv -728 \pmod{58}$

$\equiv 26 \mod 58$

$-728 + 58\times 13 = 26$

⑦ $70x \equiv 41 \mod 61$ $(0 \leq x \leq 60)$

$70 \Rightarrow 61\times 1 + \underline{9}$

$9x \equiv 41 \mod 61$

Inverse $\Rightarrow$ $(9 \mod 61)$

$\gcd(9, 61) \Rightarrow$

	$\underline{\underline{9}}$
$61 - 9\times 6 = 7$	-6
$9 - 7\times 1 = 2$	$1+6=7$
$7 - 2\times 3 = 1$	$-6-7\times 3 = -27$

$9\times -27 \equiv 1 \mod 61$

$-27 \equiv 34 \mod 61$

$61 - 27 = 34$

back to $9x \equiv 41 \mod 61 \Rightarrow$

$\times 34 \Rightarrow$ $x \equiv 41\times 34 \pmod{61}$

$x \equiv 1394 \pmod{61}$

$x = 52$

$1394 - 61\times 22 = 52$

let's verify $\Rightarrow$

$70x \equiv 41 \mod 61$

$70\times 52 = 3640$

$\equiv 41 \mod 61$ //

$3640 \div 61 = 59.67\ldots$

$3640 = 61\times 59 + 41$

(8) $13x \equiv 4 \mod 8, \quad (0 \le x \le 7)$

$13 \Rightarrow 8 \times 1 + 5$

$5x \equiv 4 \mod 8$

inverse $\Rightarrow$ $(5 \mod 8)$

gcd $(5, 8) \Rightarrow$

	$\underline{\underline{5}}$
$8 - 5 \times 1 = 3$	-1
$5 - 3 \times 1 = 2$	$1 + 1 = 2$
$3 - 2 \times 1 = 1$	$-1 - 2 = \underline{-3.}$ inverse

$5 \times -3 \equiv 1 \mod 8$ | $-3 + 8 = +5$

$\Rightarrow 5 \pmod 8$

back to $5x \equiv 4 \mod 8 \Rightarrow$

$\times 5 \Rightarrow \quad x \equiv 5 \times 4 \mod 8$

$x \equiv 20 \mod 8$

$x = 4$

let's verify $\Rightarrow$ $13x \equiv 4 \mod 8$

$13 \times 4 = 52 \equiv 4 \mod 8$